Ugly Bugs

We will use an imaginary organism call the ‘ugly bug’ to see how features of an organism can be transmitted to its offspring. We call this inheritance.

We will be coding each ugly bug using letters for each feature. We call this code an organism’s **genotype**. We will always write our code in the same order to avoid confusion later. We will write the code into boxes which will represent **genes** on **chromosomes**.

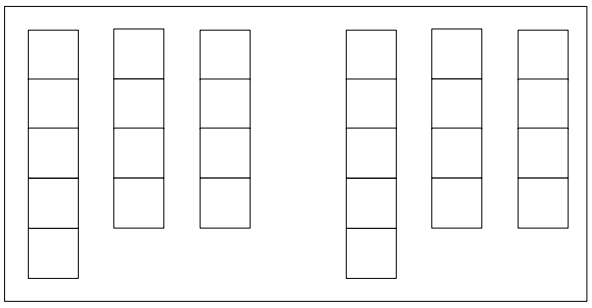
The ugly bug has 3 pairs of chromosomes in the nucleus of each of its cells. The genes and the features they control are:

Chromosome	Phenotype	Gene code		Phenotype
1.	Round head Large eyes Black eyes Long Antenna	A B C D	a b c d	Triangular head Small eyes Red eyes Short antenna
2.	Rectangular thorax Large thorax Stripes	E F G	e f g	Oval thorax Small thorax No stripes
3.	Large wings Coloured wings Hairy legs	H I J	h i j	Small wings Wings not coloured Hairless legs

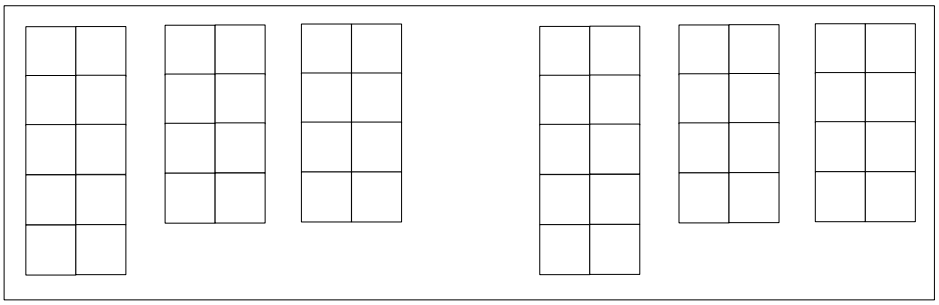
features of an organism are its **phenotype**.

Before we can begin looking at inheritance in ugly bugs, we need some ugly bugs. You need to choose head, leg, wings and body features from the possible combinations. Use the templates provided to draw your ugly bug. You will need to colour the eyes, body stripes and wings if you chose body stripes and coloured wings.

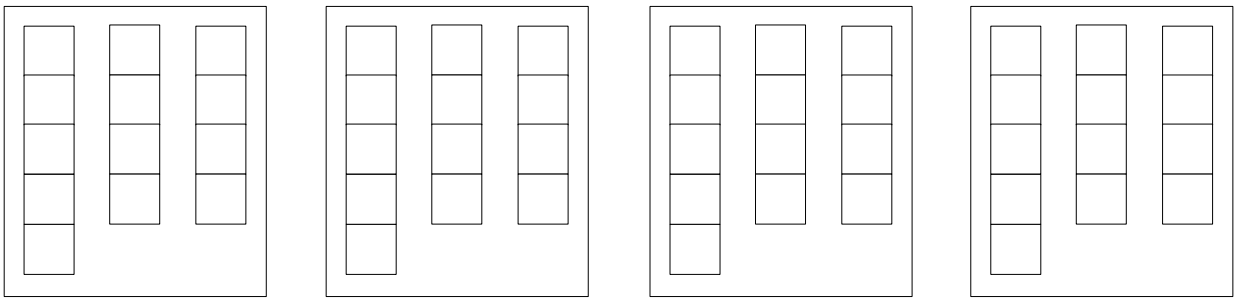
Now fill in the **gene code** for your ugly bug, using the tables of genes above as a guide. This time we will make both genes in each pair the same. Your bug is **pure bred** or **homozygous**. Now you bug is going to reproduce.



Ugly bugs reproduce sexually so they produce **gametes** by **meiosis**. Let us go through a shortened process of meiosis with the chromosomes from your ugly bug.
Step 1: The chromosomes duplicate themselves.



Step 2: After some further steps the cell divides twice to give four gametes – each gamete contains one chromosome from each pair.



Now your ugly bug is ready to reproduce. Remember that in sexual reproduction, each parent provides a gamete. The two gametes combine to make a fertilised egg or zygote. This then divides by mitosis to grow into the new organism.

Take the gene code from one of your gametes and copy it into the first line ofchromosomes.

Now copy the gene code from a gamete of a friend's ugly bug into the second line of chromosomes.

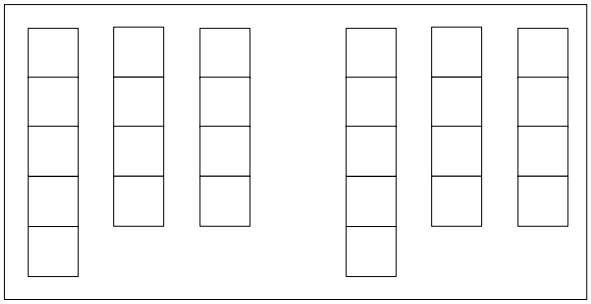
Your two ugly bugs are reproducing sexually.

Your ugly bug's gene code:

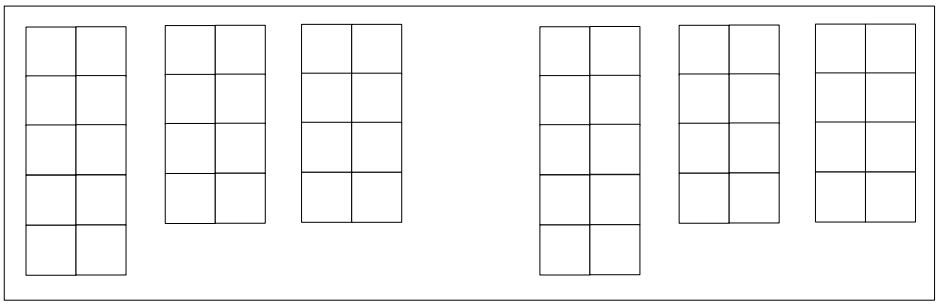
Your friend's bug's gene code:

You now have the code for a new ugly bug. From the gene code, assemble a picture of your new ugly bug. Remember that both genes play a part in deciding the features of the organism. If you have two different genes for the same feature, use the table on page 1 to decide which trait is shown.

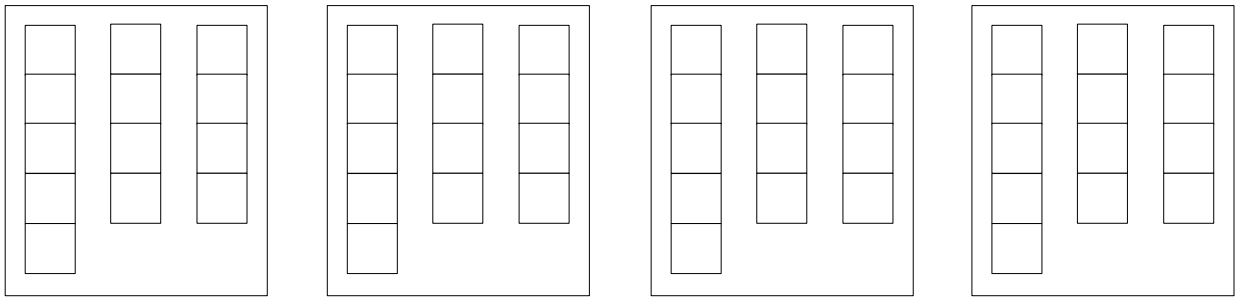
Your new ugly bug (generation 1) can now produce its own offspring. Write the gene code for your generation 1 bug below.



Step 1: The chromosomes duplicate.



Step 2: The cell divides to give four gametes.



Step 3: Choose one of the four gametes at random and write it below. Then write the gene code for a friend's ugly bug below that.

Your ugly bug's gene code:

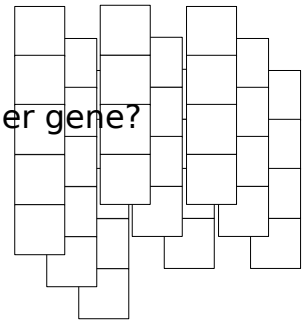
Your friend's bug's gene code:

You now have the code for a new ugly bug. From the gene code, assemble a picture of your new ugly bug. This is the second generation.

Were there any features in your second generation offspring which didn't appear in the first generation?

Why wasn't it shown in the first generation of bugs?

What do we call genes which are masked by another gene?



In summary:

Organisms which reproduce sexually produce reproductive cells called _____ which contain one set of _____. When they reproduce, each parent contributes a _____ to produce a fertilised egg or _____ , which contains all of the chromosomes from each parent. The _____ on each chromosome interact to determine the _____ of the new organism or offspring.

Some genes in the pair mask others and are called _____. The feature coded by these genes is the feature which is seen in the offspring. The feature which is not seen when this happens is said to be _____.

When an organism has two identical genes for the same feature it is said to be a _____ or _____ feature. If it has two different genes for the same feature, it is said to be a _____ or _____.

An organism may be a pure strain in some features and a hybrid in others. You may have noticed this with your ugly bug.

In real life, the genes are not always dominant or recessive. Sometimes they combine to produce an intermediate variation of a feature. We call this _____ _____.

Ugly bug features

Round head	or	Triangular head
Large eyes	or	Small eyes
Black eyes	or	Red eyes
Long antenna	or	Short antenna
Rectangular thorax	or	Oval thorax
Large thorax	or	Small thorax
Stripes	or	No stripes
Large wings	or	Small wings
Coloured wings	or	Wings not coloured
Hairy legs	or	Hairless legs